

PRINTER STATUS DASHBOARD

Hardware Architecture & LED Status System

Engineering Design Document

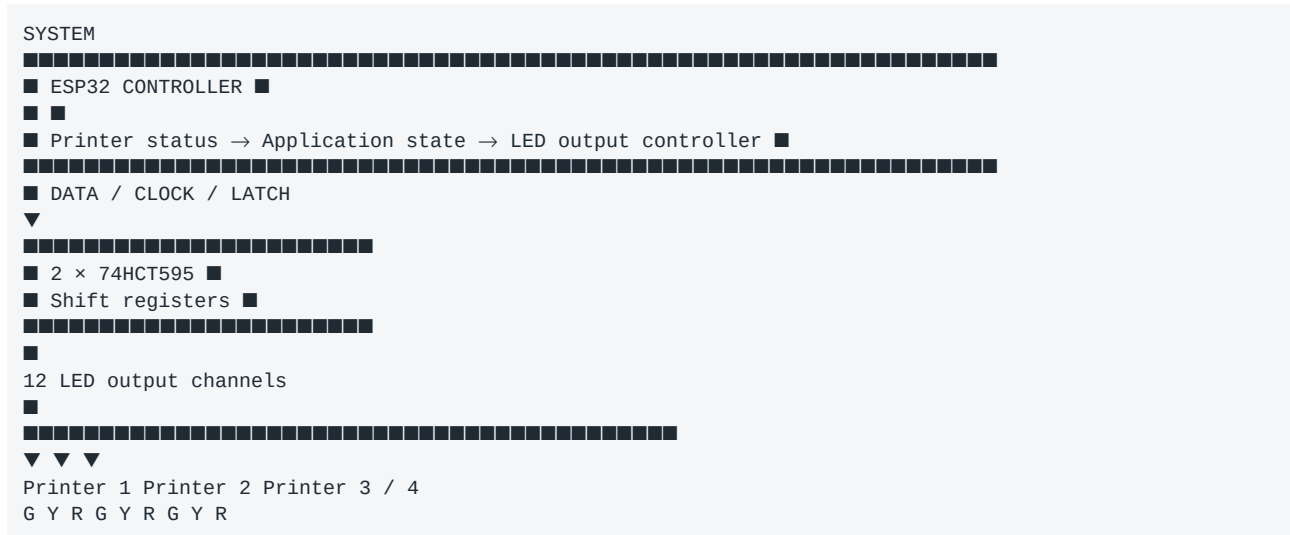
System	ESP32-based desktop printer status dashboard
Target	Four Creality-compatible network printers
Logic	3.3 V ESP32 control logic
LED Supply	5 V
Indicators	3 mm diffused LEDs
Status	Hardware architecture baseline

This document captures the hardware decisions made so far and serves as the baseline for schematic capture and PCB design.

1. System Overview

The project is a desktop dashboard for monitoring up to four network-connected 3D printers. The ESP32 is the main controller. Printer status is obtained through the CrealityKE software layer, while the hardware provides a compact visual status indication for each printer.

The LED subsystem uses three 3 mm indicator LEDs per printer. Only one status indication is active at a time for a given printer.



2. Printer Status Indication

The original five-LED concept was reduced to three LEDs to simplify the electronics and reduce the required physical indicators.

Final visual states:

State	LED	Behavior
Idle / Ready	None	All LEDs OFF
Printing	Green	Solid ON
Paused	Yellow	Solid ON
Fault / Disconnected	Red	Solid ON
Finished	Green	Blinking

Fault has the highest visual priority. If an error is detected or the printer connection is lost, the red LED takes precedence over normal operating states.

Finished is represented by a blinking green LED instead of a separate physical LED.

3. Status Encoding

Four primary logical states can be represented with two bits. Finished is a timing behavior of the green LED rather than a separate binary state.

A	B	State	LED output
0	0	Idle / Ready	OFF

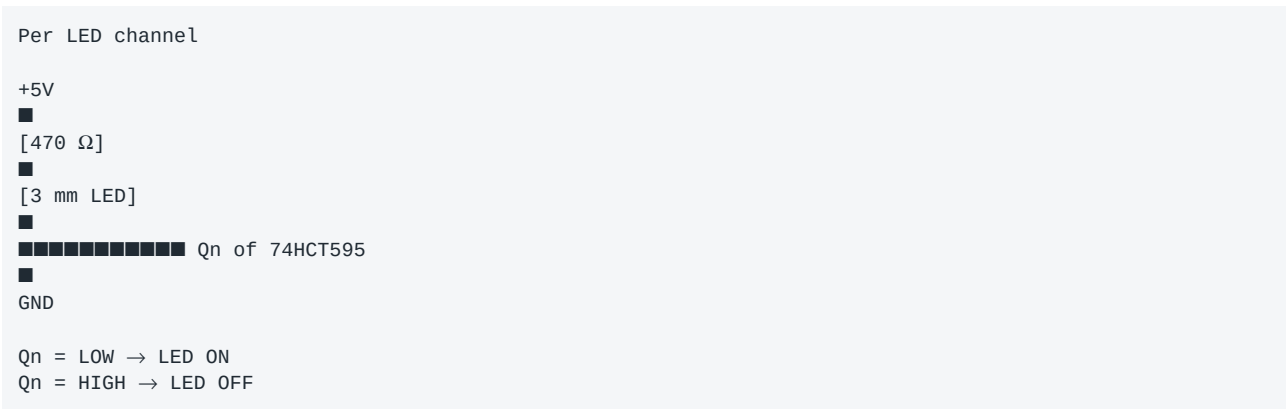
0	1	Printing	Green
1	0	Paused	Yellow
1	1	Fault	Red

The two-bit model is the logical status model. The physical implementation uses shift-register outputs, so the ESP32 does not require one GPIO per LED.

4. LED Hardware

Each printer uses three 3 mm diffused indicator LEDs: green, yellow, and red. The LEDs are powered from the 5 V rail.

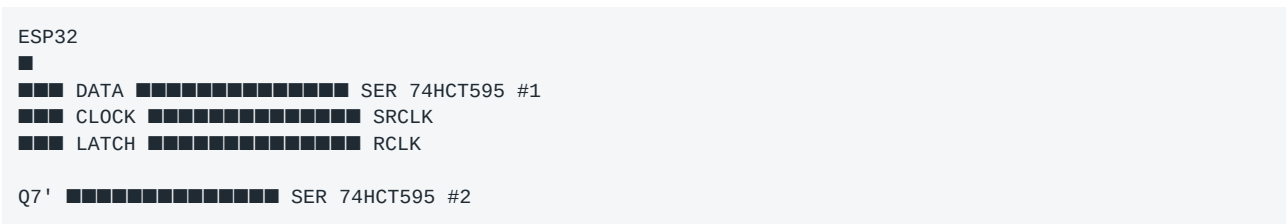
A nominal 470 Ω series resistor is assigned to every LED. With a typical red LED forward voltage near 2 V, the expected current is approximately 5–6 mA when the output is sinking the current.



The selected topology uses the 74HCT595 outputs as current sinks. No separate transistor is currently planned for each indicator.

5. Shift-Register Architecture

Two 74HCT595 8-bit serial-in/parallel-out shift registers are daisy-chained. Twelve of the sixteen available outputs are assigned to the twelve LEDs.



Output allocation:

Output	Function
Q0 #1	Printer 1 — Green
Q1 #1	Printer 1 — Yellow
Q2 #1	Printer 1 — Red
Q3 #1	Printer 2 — Green
Q4 #1	Printer 2 — Yellow
Q5 #1	Printer 2 — Red

Q6 #1	Printer 3 — Green
Q7 #1	Printer 3 — Yellow
Q0 #2	Printer 3 — Red
Q1 #2	Printer 4 — Green
Q2 #2	Printer 4 — Yellow
Q3 #2	Printer 4 — Red

Four outputs remain unused and are reserved for future expansion or additional indicators.

6. Electrical Interface & Component Selection

The 74HCT595 was selected instead of a standard 74HC595 because the controller uses 3.3 V ESP32 logic while the LED rail is 5 V. The HCT input thresholds are appropriate for the ESP32 logic level when the shift register is powered from 5 V.

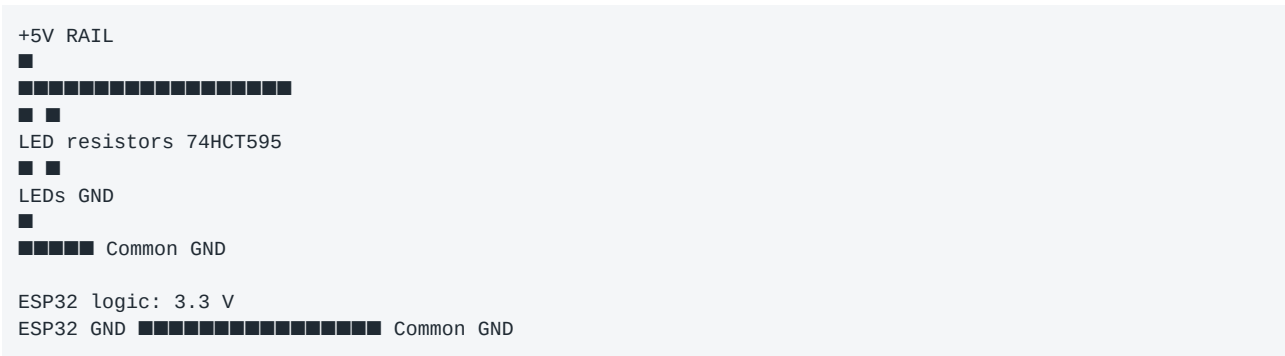
The supplied NXP datasheet specifies the relevant HCT timing/electrical operating range at 4.5–5.5 V and distinguishes recommended electrical characteristics from absolute maximum ratings. The absolute maximum output-current figures are limits, not design targets. [filecite turn1file3 L303-L326](#)

The design therefore targets approximately 5–6 mA per LED. Because only one LED is active per printer, the expected maximum simultaneous LED load is four LEDs, or roughly 20–25 mA total.

Current baseline: ULN2803A is not included. The direct current-sink topology is intentionally kept simple. Before PCB release, the exact DC electrical-characteristics table for the selected 74HCT595 manufacturer/package should be checked against the final LED current and thermal conditions.

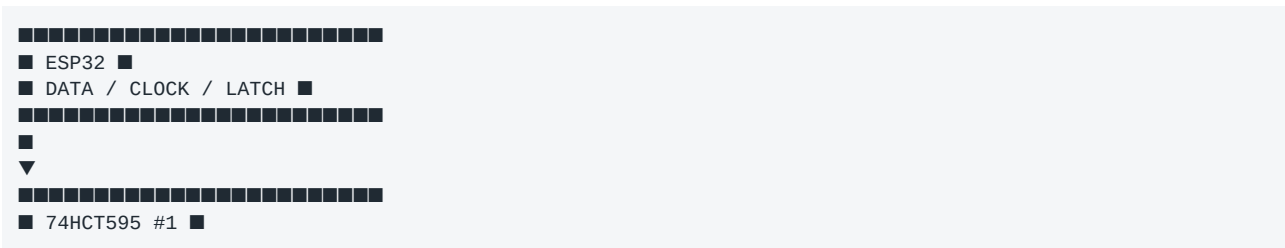
7. Power Architecture

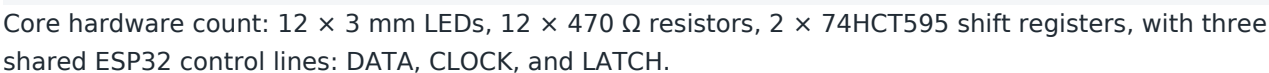
The current concept uses a 5 V rail for the LEDs and shift registers, while the ESP32 remains in its 3.3 V logic domain. ESP32 GPIOs carry logic signals only; LED current is supplied by the 5 V rail.



Local decoupling capacitors for each shift register and suitable bulk bypassing on the 5 V rail are required in the final schematic. Exact capacitor values and placement are intentionally left for the next design step.

8. Four-Printer Hardware Block





Current baseline decisions:

- ## Next hardware-design step:

This document is a design baseline, not a manufacturing release. Exact part numbers, packages, connectors, power-budget verification, and PCB routing rules remain to be finalized.